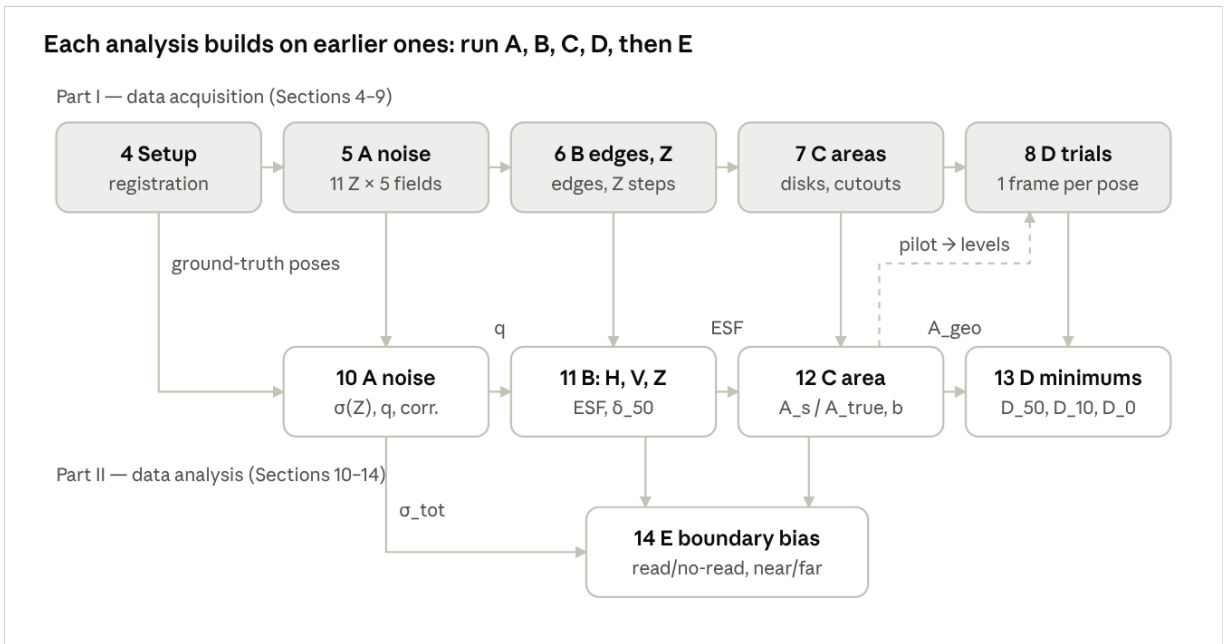


1. Purpose and scope

This procedure collects the data needed to measure five properties of the VSX3000 active binocular stereo sensor over 500–1000 mm. The sensor stays fixed. A robot carries the target, so each target pose is commanded, repeatable, and logged.

The four measurements are not independent. Procedure A supplies the noise level σ_{tot} that the boundary-bias analysis E and the warm-up and flatness gates use; B-Z and D set their detection thresholds from their own null data (the A $\rightarrow$ A visits of the Z series and the blank sites of the arrays). Procedure B supplies the edge spread function that predicts the area bias measured in C. Part I (acquisition) runs in the order A, B, C, D. Part II (analysis) runs in the same order, then E. E reuses data from B and C and needs no separate captures.



Procedure order and data dependencies · 6 capture series, 6 analyses

Shaded boxes are captures and open boxes are analyses. Registration poses feed every analysis, and the quick-look on C data sets the D trial levels (dashed).

Coordinate convention. Z runs along the left IR camera's optical axis, with its origin at that camera's optical center. H runs along image columns and V along image rows. H is assumed parallel to the

stereo baseline, with the right camera at +H; Step 4.5 checks the direction and the side. The depth image is assumed to be registered to the left IR image; Step 4.5 checks this as well.

ID	Measurand	Definition used in this procedure	Primary outputs
A	Noise vs Z	Temporal, fixed-pattern, and total depth deviation on a flat matte plate. Also fill rate, spatial correlation length, and depth quantization step.	$\sigma_t(Z)$, $\sigma_{fp}(Z)$, $\sigma_{tot}(Z)$, bias(Z), fitted disparity noise σ_d , quantum q, correlation length
B-HV	Effective resolution in H and V	The 10–90% rise distance of the depth edge spread function (ESF): the lateral distance over which the normalized depth across a knife edge rises from 10% to 90% of the step height. Also MTF50. For both edge polarities and both gaps	Rise distance in mm and px for each Z, orientation, polarity, and gap
B-Z	Effective resolution in Z	Smallest commanded Z step detected with 50% probability, for 1-px, 5×5 px, and 20×20 px patches. Also the staircase quantum and step-response gain.	$\delta_{50}(Z, \text{patch})$, $\delta Z_q(Z)$, gain
C	True vs sensed area	Sensed area at the half-height iso-contour compared with true area and with geometrically visible area, for disks and for cutouts	A_{sensed}/A_{true} vs $D/p(Z)$; edge bias $b(Z)$
D	Minimum detectable size	Diameter and subtended angle at 50%, 10%, and 0% detection probability, corrected for false alarms, for disks and for cutouts	D_{50} , D_{10} , D_0 and θ_{50} , θ_{10} , θ_0 for each Z, with confidence intervals
E	Boundary detection bias	Near a true edge, how often the sensor reports a value versus a no-read, and whether it favors the near or the far surface	Read/no-read bias index, near/far preference, and outcome-probability profiles vs distance to the edge

The outputs map onto the Tier-A forward-noise simulator parameters: σ_d maps to σ_j , the quantum to q, the measured $k = f \cdot B$ to k, and the correlation length to the IDW neighborhood N; the boundary widths W_{fab} and W_{drop} and the near/far preference π_{near} of E map to the simulator's boundary terms. Out of scope: surface-material dependence, ambient sunlight, multipath, and motion blur. Each of these can be added later as a repeat of A–E with one variable changed.